ROUXHANA FIRTHOUS

B.Tech IT • Class of 2027

Aspiring Associate Product Manager (APM) & Product / Data Analyst

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rouxhana.github.io/Portfolio

PROFESSIONAL SUMMARY

- **Profile & Engineering Foundation:** B.Tech in Information Technology undergraduate (Class of 2027) combining strong computer science foundations with dedicated focus on Product Management and Data Analytics.
- **Product Lifecycle & Roadmaps:** Proven track record steering the end-to-end **product lifecycle**, defining data-backed **product roadmaps**, authoring PRDs, and directing **cross-functional teams** across Agile sprints.
- **User Research & Analytics:** Skilled in synthesizing qualitative **user feedback**, conducting empirical **market research**, instrumenting product **KPIs** via SQL and Python, and aligning **stakeholder management** with customer outcomes.
- **Modern Tooling:** Passionate about leveraging cutting-edge **Agentic AI** tools and cloud architectures to accelerate product delivery and user value.

SKILLS

- **Product Management & Strategy:** Product Roadmaps, PRD Authoring, Agile & Scrum Methodologies, Stakeholder Management, User Stories & Acceptance Criteria, Feature Prioritization (RICE, MoSCoW), User Journey Mapping, User Feedback Loops, Product Lifecycle Management (PLM), Market Research.
- **Data Analytics & Product KPIs:** SQL (Complex Joins, Aggregations, Funnel Analysis), Power BI & Tableau Dashboards, Product Health & Retention KPIs, Cohort Tracking, Excel / Google Sheets Financial Modeling, Python (Pandas, NumPy, Exploratory Data Analysis), A/B Testing Principles.
- **AI, Cloud & Engineering:** Agentic AI & Autonomous LLM Systems (Autonomous Agents, Tool Calling, Prompt Orchestration), AWS Cloud Foundations (EC2, S3), HTML5, CSS3, JavaScript, Java Programming, React Fundamentals, Git & GitHub Workflows, JIRA & Confluence Workflows, High-Speed Typewriting.

PROFESSIONAL EXPERIENCE

Student Associate Product Manager (APM) • Technical Vertical Lead

Aug 2024 – Present

Department Innovation & Learning Vertical • Tamil Nadu, India

- **Product Roadmap & Sprints:** Defined academic product roadmap and directed sprint planning across cross-functional teams of 12+ student engineers, accelerating feature delivery velocity by 25%.
- **Stakeholder Management:** Spearheaded ongoing stakeholder management with faculty mentors, aligning curriculum goals with student learning milestones.
- **User Feedback & KPIs:** Synthesized qualitative user feedback from 300+ students and instrumented engagement KPIs, driving a +38 Net Promoter Score (NPS) lift.
- **PRD Authoring:** Executed competitive market research across peer programs; authored 8+ structured PRDs with acceptance criteria, user stories, and system flowcharts.

Product Analyst & Core Contributor • AapThozhi Initiative

Jan 2024 – Aug 2024

CivicTech Open Source Team • Hybrid / Remote

- **Product Lifecycle Management:** Managed the end-to-end product lifecycle from user discovery through prototype testing for a voice-first government welfare navigator across 14 Indian languages.

- **Voice-First Navigation:** Increased user engagement by 34% by integrating Web Speech API and accessible voice navigation for low-literacy audiences.
- **User Research:** Conducted empathy-driven user feedback interviews with 40+ grassroots participants to identify drop-off friction points.
- **Funnel Optimization & Prioritization:** Tracked scheme discovery funnels via SQL and Python; applied RICE prioritization to eliminate 3 verification hurdles, reducing application form abandonment by 42%.

FEATURED PRODUCT & ENGINEERING PROJECTS

AapThozhi — Multilingual Civic Welfare Support Platform

`React 19 • TypeScript • Web Speech API • CivicTech`

Conceived and developed an accessible voice-assisted welfare portal addressing digital literacy barriers for government scheme discovery across 14 Indian languages.

- Authored full product specifications (PRDs), personas, and user journeys based on demographic research, attaining a 99.2% scheme qualification accuracy rate in simulated testing.
- Engineered speech-to-text navigation with Web Speech API and counter-ready verification scripts, lowering user onboarding friction by 40% and decreasing application form abandonment by 42%.
- Optimized local state caching and speech synthesis latency to under 350ms, maintaining 98% positive user feedback across low-bandwidth 2G/3G mobile networks.

AgriHelp (Agri-Guide-Mini) — AI Agricultural Guidance System

`JavaScript • AI Assistant • UX Research • AgriTech`

Designed and launched a data-driven agricultural decision-support web platform tailored for smallholder farmers in Tamil Nadu with under 1 acre of land.

- Conducted field market research with 50+ local smallholder farmers, mapping cultivation constraints and translating seasonal volatility into a 3-stage crop roadmap.
- Developed dynamic crop recommendation algorithm based on soil N-P-K parameters and market price fluctuations, reducing preliminary crop selection overhead by 28%.
- Defined product health KPIs and instrumented responsive mobile UI, lowering query resolution time to under 1.8 seconds and increasing multi-session retention by 22%.

Product Portfolio & Interactive Casebook

`HTML5 • Tailwind CSS • Vanilla JS • ATS & SEO Optimized`

Engineered an editorial, full-screen portfolio and automated ATS-compliant PDF resume pipeline achieving 100/100 Lighthouse performance and SEO scores.

- Structured interactive data dashboards showcasing SQL joins, Power BI models, and Agile product frameworks with direct GitHub traceability.
- Implemented responsive CSS architecture ensuring 100% full-screen coverage from mobile (320px) to ultra-wide displays (2560px+).

EDUCATION

B.Tech in Information Technology

`2023 - 2027`

Tamil Nadu, India

Key Coursework: Agentic AI Systems, Cloud Computing (AWS), Data Structures & Algorithms, Database Management Systems (SQL), Software Engineering & Agile Methods, Web Technologies.

LEADERSHIP & CERTIFICATIONS

Agentic AI & Autonomous Systems

`Coursework`

College-accredited coursework covering autonomous agent architectures, multi-agent coordination, tool-augmented LLMs, and evaluation frameworks.

AWS Cloud Basics Course

`Certified`

Accredited coursework covering AWS EC2 compute, S3 storage, IAM security policies, and cloud scalability.

Learning Vertical Chair — Elected

Lead department technical learning workshops, peer mentorship circles, and technology translation webinars.

Professional Typewriting Certification — Certified

State-certified high-speed typewriting qualification supporting rapid documentation and PRD authoring.